

NEWCASTLE WEATHER CENTRE, United Kingdom

WMO: 032460

Lat: 54.983N

Lon: 1.600W

Elev: 171

StdP: 14.61

Time Zone: 0.00 (EUW)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	28.5	30.5	20.2	15.3	33.6	22.5	17.0	34.8	24.6	43.4	21.5	43.2	5.2	270	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.5	74.1	62.6	71.4	61.7	68.9	59.9	64.4	71.8	62.8	69.0	61.4	67.1	7.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.6	82.5	66.5	60.2	78.4	65.8	58.6	74.0	64.7	29.5	71.9	28.4	69.1	27.4	67.1	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.3	17.9	15.9	DB	23.0	80.5	3.4	2.7	20.5	82.4	18.5	84.0	16.6	85.5	14.1	87.5
			WB	22.0	68.4	3.2	2.8	19.7	70.4	17.8	72.0	16.1	73.6	13.8	75.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	41.1	41.7	43.9	46.7	51.6	56.6	60.5	60.9	56.6	51.1	45.8	40.9	(6)
	DBStd	8.67	4.80	5.38	5.22	4.96	5.17	4.84	4.25	4.34	3.91	4.88	4.86	5.64	(7)
	HDD50	1352	277	237	197	123	42	3	0	0	1	46	140	286	(8)
	HDD65	5568	740	652	654	548	414	256	151	143	254	432	576	748	(9)
	CDD50	1290	1	5	8	24	92	201	325	338	199	79	15	3	(10)
	CDD65	31	0	0	0	0	0	4	10	15	2	0	0	0	(11)
	CDH74	93	0	0	0	0	1	13	26	50	3	0	0	0	(12)
	CDH80	6	0	0	0	0	0	1	0	5	0	0	0	0	(13)
(14) Wind	WSAvg	7.1	7.8	8.5	7.8	7.6	7.1	7.1	6.2	6.2	6.4	7.0	6.7	7.1	(14)
(15) Precipitation	PrecAvg	27.2	2.1	1.9	1.6	2.1	2.0	2.3	2.6	2.6	2.2	2.4	2.9	2.4	(15)
	PrecMax	39.6	5.0	3.9	3.4	5.6	3.9	6.5	6.6	6.6	5.0	4.6	5.9	5.3	(16)
	PrecMin	19.1	0.5	0.3	0.4	0.3	0.6	0.5	0.4	0.5	0.6	0.4	0.8	0.9	(17)
	PrecStd	4.3	1.1	1.0	0.9	1.6	0.9	1.5	1.3	1.3	1.3	1.0	1.4	1.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	57.3	58.5	64.7	71.1	75.4	77.4	79.6	74.0	66.3	59.4	58.3	(19)
		MCWB	50.3	51.9	50.1	54.6	59.7	62.6	64.1	64.4	62.1	57.0	54.4	53.5	(20)
	2%	DB	52.6	54.2	56.1	59.7	66.4	70.2	73.4	74.4	69.4	62.8	56.9	53.7	(21)
		MCWB	48.2	49.3	49.2	51.1	56.6	59.5	63.0	62.9	59.5	56.3	52.4	49.3	(22)
	5%	DB	50.4	52.1	54.2	57.2	62.9	67.6	70.8	71.3	65.6	60.5	54.7	51.4	(23)
		MCWB	46.4	47.4	48.1	49.9	53.6	57.7	61.8	62.1	57.5	54.8	50.7	47.3	(24)
	10%	DB	48.5	50.0	52.1	54.8	60.2	64.8	68.1	68.3	63.2	58.4	53.0	48.6	(25)
		MCWB	44.9	45.4	46.7	48.8	52.4	56.2	59.6	60.4	56.3	53.6	49.5	45.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.7	52.2	52.9	55.3	61.7	65.2	67.0	66.3	63.9	58.7	54.8	53.7	(27)
		MCDB	54.5	56.7	57.3	62.8	69.0	72.6	73.5	73.4	71.5	64.0	58.2	58.5	(28)
	2%	WB	48.5	50.1	50.4	52.8	57.6	61.3	64.3	64.3	61.2	57.2	53.1	49.8	(29)
		MCDB	51.7	54.0	54.8	58.6	65.1	68.4	71.3	71.9	66.7	61.9	56.2	52.7	(30)
	5%	WB	47.0	47.7	48.8	51.0	55.0	59.1	62.3	63.0	59.3	55.6	51.6	47.9	(31)
		MCDB	50.0	51.7	53.1	55.8	61.4	65.6	68.6	69.0	63.3	59.3	54.0	50.8	(32)
	10%	WB	45.3	45.8	47.3	49.2	53.1	57.1	60.8	61.8	57.5	54.1	50.1	45.8	(33)
		MCDB	48.3	49.6	51.5	54.0	58.8	63.3	67.0	66.9	61.7	57.9	52.5	48.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.7	9.7	10.7	11.4	11.0	11.3	11.5	10.8	9.6	7.8	7.3	(35)
		MCDBR	10.1	10.1	12.5	14.2	15.8	15.6	15.7	15.8	15.5	11.7	9.2	9.6	(36)
	5% WB	MCWBR	9.0	8.6	8.1	8.3	8.5	8.3	8.0	7.1	8.2	8.0	8.0	8.5	(37)
		MCWBR	9.7	9.9	11.0	13.6	15.1	14.4	14.1	13.9	13.1	11.1	8.5	9.5	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.317	0.334	0.366	0.390	0.387	0.389	0.390	0.380	0.360	0.344	0.318	0.301	(40)
		taud	2.312	2.379	2.308	2.267	2.324	2.345	2.348	2.390	2.448	2.450	2.370	2.280	(41)
	Ebn at Noon	Ebn at Noon	193	237	254	264	272	272	269	265	255	231	193	171	(42)
		Edh at Noon	18	23	31	37	37	37	36	33	27	22	16	15	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	180	381	742	1204	1559	1562	1497	1242	884	481	227	134	(44)
	RadStd	RadStd	23	42	91	117	86	128	121	82	75	41	20	14	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page